

Suggested Solution

1 (a)

$$\mu = \frac{ve}{F}$$

$$[\mu] = \frac{[v][e]}{[F]}$$

$$= \frac{\text{m s}^{-1} (\text{C})}{\text{N}} \quad [\text{M1}]$$

$$= \frac{\text{m s}^{-1} (\text{A s})}{\text{kg m s}^{-2}}$$

$$= \text{A kg}^{-1} \text{s}^2 \quad [\text{A1}]$$

-1 if use C and/or N

-1 if poor presentation

(b) (i) Zero error or wrongly calibrated scale [B1]

(ii) Reading scale from different angles [B1]

Must explain if write parallax error.

$$(c) \quad (i) \quad R_x = \frac{V}{I} = \frac{3.00}{4.9 \times 10^{-3}} \quad [\text{B1}]$$

$$= 612 \, \Omega$$

$$(ii) \quad \frac{\Delta R_x}{R_x} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R_x}{612} = \frac{0.03}{3.00} + \frac{0.1}{4.9}$$

$$\Delta R_x = 20 \, \Omega \quad [\text{M1}]$$

$$R_x \pm \Delta R_x = 610 \pm 20 \, \Omega \quad [\text{A1}]$$

(d) When plotting a graph, a best fit line shows the average trend of the data points, thus reducing random error present. Those readings which are too far out from the graph are not taken into account so the results will be more reliable. [B1]

OR

By using multiple sets of values and plotting them on a graph, any consistent deviation from the expected trend can be observed. If all points lie away from the expected line or curve in a similar manner, this may indicate the presence of a systematic error.

OR

When several data points are collected and plotted, it becomes easier to identify anomalous readings (readings that deviate significantly from the



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general pattern or trend). These readings are not taken into account so the mean value of resistance of X will be closer to the actual value..